of the stress test, particularly the stage of exercise at which ECG signs of ischemia appear; the magnitude and number of the ischemic zones of myocardium on imaging; and the change in LV ejection fraction that occurs on radionuclide ventriculography or echocardiography during ischemia and/or during exercise; (2) the ECG leads showing a positive response, with changes in the anterior precordial leads indicating a less favorable prognosis than changes in the inferior leads; and (3) the patient's age, occupation, and general medical condition.

Most would agree that an asymptomatic 45-year-old commercial airline pilot with significant (0.4-mV) ST-segment depression in leads V_1 to V_4 during mild exercise should undergo coronary arteriography, whereas an asymptomatic, sedentary 85-year-old retiree with 0.1-mV ST-segment depression in leads II and III during maximal activity need not. However, there is no consensus about the most appropriate approach in the large majority of patients for whom the situation is less extreme. Asymptomatic patients with silent ischemia, three-vessel CAD, and impaired LV function may be considered appropriate candidates for CABG.

The treatment of risk factors, particularly lipid lowering and blood pressure control as described above, and the use of aspirin, statins, and beta blockers after infarction have been shown to reduce events and improve outcomes in asymptomatic as well as symptomatic patients with ischemia and proven CAD. Although the incidence of asymptomatic ischemia can be reduced by treatment with beta blockers, calcium channel blockers, and long-acting nitrates, it is not clear whether this is necessary or desirable in patients who have not had a myocardial infarction.

FURTHER READING

American Diabetes Association Professional Practice Committee: Cardiovascular disease and risk management: Standards of care in diabetes—2024. Diabetes Care 47:S179, 2024.

Antman EM, Braunwald E: Managing stable ischemic heart disease. N Engl J Med 382:1468, 2020.

Gaba PK et al: Association between achieved low-density lipoprotein cholesterol levels and long-term cardiovascular and safety outcomes: An analysis of FOURIER-OLE. Circulation 147:1192, 2023.

Gulati M et al: 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 144:e368, 2021.

Hokimoto S et al: JCS/CVIT/JCC 2023 guideline focused update on diagnosis and treatment of vasospastic angina (coronary spastic angina) and coronary microvascular dysfunction. Circ J 87:879, 2023.

Knuuti J et al: 2019 ESC guidelines for the diagnosis and management of chronic coronary syndromes. Eur Heart J 41:407, 2020.

Lloyd-Jones DM et al: 2022 ACC expert consensus decision pathway on the role of nonstatin therapies for LDL-cholesterol lowering in the management of atherosclerotic cardiovascular disease risk. J Am Coll Cardiol 80:1366, 2022.

Maron DJ et al: Initial invasive or conservative strategy for stable coronary disease. N Engl J Med 382:1395, 2020.

Martin SS et al: 2024 Heart disease and stroke statistics: A report of US and global data from the American Heart Association. Circulation 149:e347, 2024.

Marx N et al: 2023 ESC Guidelines for the management of cardiovascular disease in patients with diabetes. Eur Heart J 44:4043, 2023.

Mensah GA et al: Global burden of cardiovascular diseases and risks, 1990-2022. J Am Coll Cardiol 82:2350, 2023.

Rao SV et al: Chronic coronary disease guidelines. Circulation 148:729, 2023.

Virani SS et al: 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients With Chronic Coronary Disease: A report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines [published correction appears in Circulation 148:e148, 2023] [published correction appears in Circulation 148:e186, 2023]. Circulation 148:e9, 2023.

285 Non-ST-Segment Elevation Acute Coronary Syndrome (Non-ST-Segment Elevation Myocardial Infarction and Unstable Angina)

Robert P. Giugliano, Christopher P. Cannon, Eugene Braunwald

Patients with acute coronary syndrome (ACS) are commonly classified into two groups to facilitate evaluation and management, namely patients with acute myocardial infarction (MI) with ST-segment elevation (STEMI) on their presenting electrocardiogram (ECG) **(see Chap. 286)** and those with non-ST-segment elevation acute coronary syndrome (NSTE-ACS). The latter include patients with non-ST-segment elevation MI (NSTEMI), who, by definition, have evidence of myocyte necrosis, and those with unstable angina (UA), who do not **(Fig. 285-1)**.

The incidence of NSTEMI is rising due to the increasing burden of obesity, diabetes, and chronic kidney disease in an aging population and the increasing detection of myocardial necrosis by troponin (see below), whereas the incidence of STEMI is declining due to greater use of aspirin, statins, and less smoking. Among patients with NSTE-ACS, the proportion with NSTEMI is increasing while that with UA is falling because of the wider use of highly sensitive troponin (hsTn) assays (see below) with enhanced detection of myocyte necrosis, thereby reclassifying UA to NSTEMI.

PATHOPHYSIOLOGY

NSTE-ACS is caused by an imbalance between myocardial oxygen supply and demand, most commonly resulting from one or more of three processes that lead to coronary arterial thrombosis: (1) plaque fissure with inflammation—the inflammatory response is reflected by an increased activity of effector T cells as part of an adaptive immunity dysregulation; (2) plaque fissure without inflammation; and (3) plaque erosion, which is present in at least one-third of ACS and is recognized with increasing frequency **(Fig. 285-2)**. The so-called "vulnerable plaques" responsible for ACS may show an eccentric stenosis with scalloped or overhanging edges and a narrow neck on coronary angiography. Such plaques usually are composed of a lipid-rich core with a thin fibrous cap. Patients with NSTE-ACS frequently have multiple such plaques that are at risk of disruption. A fourth process, without thrombosis, may be caused by epicardial or microvascular spasm or increased myocardial oxygen demand, typically in the presence of fixed epicardial coronary obstruction.

Among patients with NSTE-ACS studied at angiography, <10% have stenosis of the left main coronary artery, and approximately 35% have three-vessel coronary artery disease, 25% have two-vessel disease, 20% have single-vessel disease, and 10% have no apparent critical epicardial coronary artery stenosis; some of the latter may have obstruction of the coronary microcirculation and/or spasm of the epicardial vessels.

CLINICAL PRESENTATION

Diagnosis The diagnosis of NSTE-ACS is based largely on a combination of the history and clinical findings (age, ECG, and circulating troponin) **(Table 285-1, Fig. 285-3)**.

History and Physical Examination Typically, chest discomfort is severe and has at least one of three features: (1) occurrence at rest (or with minimal exertion), lasting >10 min; (2) of relatively recent onset (i.e., within the prior 2 weeks); and/or (3) a crescendo pattern, i.e., distinctly more severe, prolonged, or frequent than previous episodes.

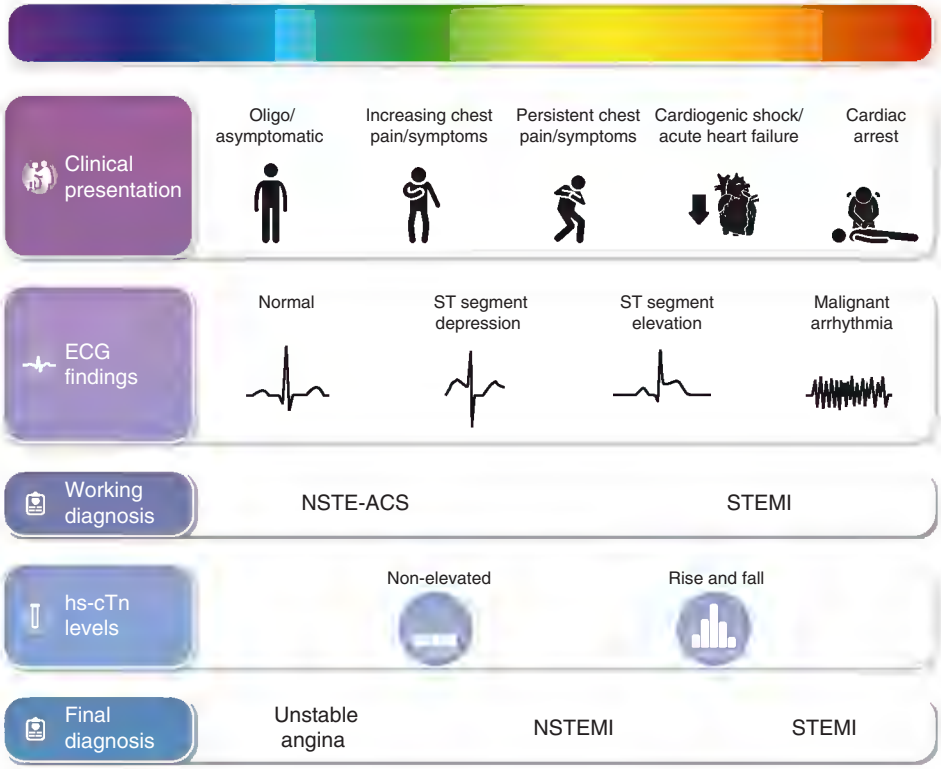

FIGURE 285-1 Spectrum of acute coronary syndromes. The spectrum of clinical presentations, electrocardiographic findings, and high-sensitivity cardiac troponin levels in patients with acute coronary syndrome. ACS, acute coronary syndrome; ECG, electrocardiogram; hs-cTn, high-sensitivity cardiac troponin; NSTE-ACS, non-ST-segment elevation acute coronary syndrome; NSTEMI, non-ST-segment elevation myocardial infarction; STEMI, ST-segment elevation myocardial infarction. *(RA Byrne et al: 2023 ESC guideline for the management of acute coronary syndromes. Eur Heart J 44:3720, 2023, by permission of Oxford university Press.)*

The diagnosis of NSTEMI is established if a patient with any of these features (without ECG ST-segment elevations) develops evidence of myocardial necrosis, as reflected in abnormally elevated levels of circulating troponin, in the absence of another explanation (see below). The chest discomfort is typically located in the substernal region and radiates to the left arm, left shoulder, and/or superiorly to the neck and jaw. Anginal equivalents such as dyspnea, epigastric discomfort, nausea, or weakness may occur instead of chest discomfort. These equivalents are more frequently reported by women, the elderly, and patients with diabetes mellitus. The physical examination resembles that in patients with stable angina **(see Chap. 284)** and may be unremarkable. However, if the patient has a large area of myocardial ischemia or a large NSTEMI, the physical findings can include diaphoresis; pale, cool skin; sinus tachycardia; a third and/or fourth heart sound; basilar rales; hypotension; and in severe cases, cardiogenic shock. An important goal of the physical examination is to identify potential alternative life-threatening diagnoses, e.g., pulmonary embolism **(see Chap. 290)**, aortic dissection **(see Chap. 291)**, and cardiac tamponade **(see Chap. 281)**,

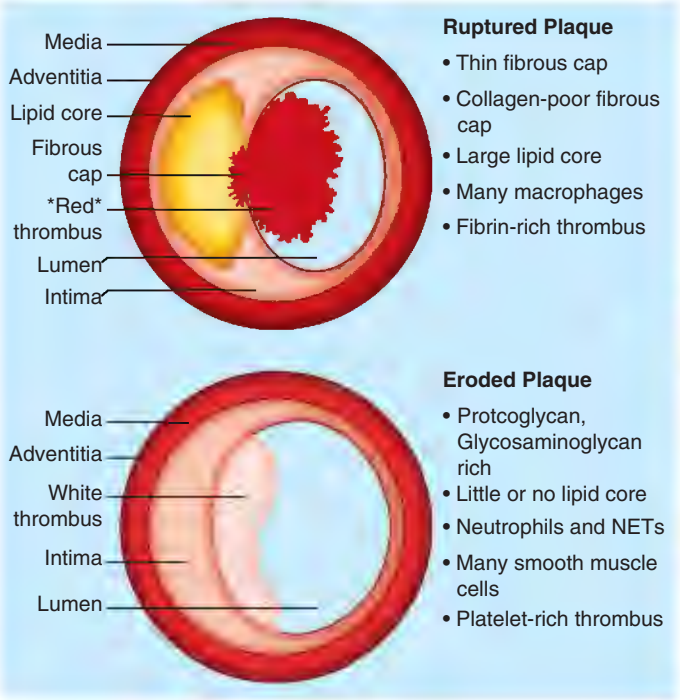

FIGURE 285-2 Comparison of the characteristics of human atheromata complicated by thrombosis and causing acute coronary syndrome. The column on the left highlights some of the characteristics demonstrated by analyses of human coronary arterial lesions that have undergone thrombosis by these two diverse mechanisms. NETs, neutrophil extracellular traps. *(Reproduced with permission from P Libby et al: Reassessing the mechanisms of acute coronary syndromes the "vulnerable plaque" and superficial erosion. Circ Res 124:150, 2019.)*

TABLE 285-1 TIMI Risk Score for NSTE-ACS

RISK MARKERS	
• Age ≥65 years • Known CAD (≥50% stenosis) • ST deviation >0.5 mm on presenting ECG	• ↑ cardiac markers • ≥2 original episodes in prior 24 h • Prior angina • ≥3 CAD risk factors
NO. OF RISK MARKERS	**INCIDENCE OF ADVERSE CARDIAC EVENTS[a] (%)**
0/1	5
2	8
3	13
4	20
5	26
6/7	41

[a]Risk at 14 days of death, new or recurrent myocardial infarction, or severe recurrent ischemia requiring urgent revascularization.

Abbreviations: CAD, coronary artery disease; ECG, electrocardiogram; NSTE-ACS, non-ST-segment elevation acute coronary syndrome; TIMI, Thrombolysis in Myocardial Infarction.

FIGURE 285-3 The HEART pathway for evaluation of acute chest pain. CAD, coronary artery disease; ECG, electrocardiogram; STEMI, ST-segment elevation myocardial infarction. *The Heart Score assigns 0, 1, or 2 points depending on the extent of abnormality for each of the History, ECG, Age, Risk factors, and Troponin (*Six AJ, Backus BE, and Kelder JC. Chest pain in the emergency room: value of the HEART score. Neth Heart J 16:191, 2008*). *(Reproduced with permission from JL Januzzi et al: Recommendations for institutions transitioning to high-sensitivity troponin testing: JACC scientific expert panel. J Am Coll Cardiol 73:1059-1077, 2019.)*

given the vastly different treatment plans required to avoid major morbidity and death.

Electrocardiogram New ST-segment depression occurs in about one-third of patients with NSTE-ACS. It may be transient but can persist for as long as several days following NSTEMI. T-wave changes are more common but are a less specific sign of ischemia unless they are new and deep T-wave inversions (≥0.3 mV).

Cardiac Biomarkers Patients with NSTEMI have elevated biomarkers of necrosis, such as cardiac troponin (cTn) I or T (cTnI or cTnT). High-sensitive (hs) cTns are sensitive, relatively specific, and the preferred markers of myocardial necrosis. Elevated levels of cTn with a dynamic early change distinguish patients with NSTEMI from those with UA. In patients with NSTEMI, there is a characteristic temporal rise of the plasma concentration, peaking at 12–24 h after onset of symptoms and gradually decreasing thereafter. There is a direct relationship between the degree of elevation and mortality. The 1-h rapid rule-out MI algorithm (no abnormal elevation of hsTn at 0 or 1 h after presentation) has been recommended by recent practice guidelines. It is important to distinguish myocardial injury from myocardial necrosis; the former is defined by elevations of cTn >99th percentile of the upper reference limit in patients *without* a clear clinical history or ECG features of acute myocardial ischemia. Myocardial injury may be caused by a variety of noncardiac and cardiac conditions other than MI (Table 285-2).

RISK STRATIFICATION

Patients with documented NSTE-ACS exhibit a wide spectrum of early (30 days) risk of death, ranging from 1 to 10%, and a recurrent ACS rate of 5–15% during the first year. Assessment of risk can be accomplished by one of several clinical risk scoring systems, including those developed from the Thrombolysis in Myocardial Infarction (TIMI) Trials, the Global Registry of Acute Coronary Event (GRACE), and the HEART (*h*istory, *e*lectrocardiogram, *a*ge, *r*isk factors, *t*roponin) score (Fig. 285-3). Multimarker strategies including hsTn and brain natriuretic peptides are now gaining favor, both to define more fully the pathophysiologic mechanisms underlying a patient's presentation and to stratify the patient's risk further. Early risk assessment is useful in identifying patients who would derive the greatest benefit from an early invasive strategy (see below).

TABLE 285-2 Reasons for the Elevation of Cardiac Troponin Values as a Result of Myocardial Injury

Myocardial injury related to acute myocardial infarction
Atherosclerotic plaque disruption or erosion with thrombosis
Myocardial injury related to acute myocardial ischemia because of oxygen supply/demand imbalance
Reduced myocardial perfusion
• Coronary artery spasm, microvascular dysfunction
• Coronary embolism
• Coronary artery dissection
• Sustained bradyarrhythmia
• Hypotension or shock
• Respiratory failure
• Severe anemia
Increased myocardial oxygen demand
• Sustained tachyarrhythmia
• Severe hypertension
Other causes of myocardial injury
Cardiac conditions
• Heart failure
• Myocarditis
• Cardiomyopathy (any type)
• Takotsubo syndrome
• Recent coronary revascularization
• Cardiac procedure other than revascularization
• Catheter ablation
• Defibrillator shocks
• Cardiac contusion
Systemic conditions
• Sepsis
• Chronic kidney disease
• Stroke, subarachnoid hemorrhage
• Pulmonary embolism
• Infiltrative diseases, e.g., amyloidosis, sarcoidosis
• Chemotherapeutic agents
• Critical illness
• Strenuous exercise

Note: For a more comprehensive listing, see the Fourth Universal Definition of Myocardial Infarction (source).

Source: Reproduced with permission from K Thygesen et al: Fourth universal definition of myocardial infarction (2018). Circulation 138:2231, 2018.

TREATMENT

Non-ST-Segment Elevation Acute Coronary Syndrome

Patients with a low likelihood of ischemia can usually be managed in an emergency department or a dedicated "chest pain unit." Evaluation of such patients includes clinical monitoring for recurrent ischemic discomfort and continuous monitoring of ECGs, serial monitoring of hsTn at presentation, 1 h, and, if necessary, 3 h. In patients for whom there is diagnostic uncertainty or intermediate-risk patients, stress testing to detect and grade ischemia or cardiac computed tomography angiography to assess epicardial coronary artery obstruction may be considered **(see Chaps. 248 and 284)**.

MEDICAL TREATMENT

Patients who "rule-in" for NSTE-ACS by clinical features, hsTn, or ST-T–wave changes on the ECG should be admitted to the hospital. Patients should be placed on bed rest with continuous ECG monitoring for ST-segment deviation and cardiac arrhythmias, preferably on a specialized cardiac unit. Ambulation is permitted if the patient shows no recurrence of ischemia (symptoms or ECG changes) and does not develop an elevation of cTn for 24 h.

Medical therapy consists of an acute phase focused on the clinical symptoms and stabilization of the culprit lesion(s) and a longer-term phase that involves therapies directed at the prevention of disease progression and future recurrent NSTE-ACS.

ANTI-ISCHEMIC TREATMENT (TABLE 285-3)

To provide relief of pain and discomfort, initial treatment, in addition to bed rest, includes nitrates, β-adrenergic blockers, and inhaled oxygen in patients with hypoxemia (arterial O_2 saturation <90%) and/or in those with heart failure and rales.

Nitrates Nitroglycerin should first be given sublingually or by buccal spray (0.3–0.6 mg) if the patient is experiencing ischemic discomfort. If symptoms persist after three doses given 5 min apart, intravenous nitroglycerin (5–10 μg/min, using nonabsorbing tubing) is recommended. The infusion rate may be increased by 10 μg/min every 3–5 min until symptoms are relieved, systolic arterial pressure falls to <90 mmHg, or the dose reaches 200 μg/min. Topical or oral nitrates **(see Chap. 284)** can be used when the pain has resolved, or they may replace intravenous nitroglycerin when the patient has been symptom-free for 12–24 h. The only absolute contraindications to the use of nitrates are hypotension or the recent use of a phosphodiesterase type 5 (PDE-5) inhibitor, sildenafil or vardenafil (within 24 h), or tadalafil (within 48 h).

β-Adrenergic Blockers and Other Agents Beta blockers are the other mainstay of anti-ischemic treatment because they reduce myocardial oxygen needs. They may be started by the intravenous route in patients with severe ischemia but should be avoided in the presence of acute or severe heart failure, low cardiac output,

TABLE 285-3 Recommendations for Anti-Ischemic Drugs in the Acute Phase of NSTE-ACS

THERAPY	RECOMMENDATION	WHEN TO AVOID
Nitrates	• Use sublingual or intravenous nitrates to relieve angina • Use intravenous nitrates if recurrent angina, uncontrolled hypertension, or signs of heart failure • Consider in patients with vasospastic angina	• Recent use of a PDE-5 inhibitor[a] • Hypotension • Right ventricular infarct • Severe aortic stenosis
Beta blockers	• Initiate early for ischemic symptoms • Continue chronic therapy	• PR interval >0.24 s • 2nd or 3rd atrioventricular block • Heart rate <50 beats/min • Systolic pressure <90 mmHg • Shock or Killip class III or IV heart failure • Severe reactive airways disease
Calcium channel blockers	• Consider in patients with vasospastic angina • Consider in patients with contraindications to beta blockers	• Systolic pressure <90 mmHg • Pulmonary edema • Left ventricular dysfunction[b] • Avoid short-acting nifedipine
Morphine or other narcotic analgesics[c]	• Continued severe angina despite 3 sublingual nitroglycerin tablets • Recurrent ischemia despite adequate anti-ischemic therapy	• Hypotension • Respiratory depression • Confusion or obtundation

[a]Sildenafil or vardenafil <24 h or tadalafil <48 h. [b]Diltiazem or verapamil.
[c]Concomitant administration may delay the absorption and blunt the antiplatelet effect of oral $P2Y_{12}$ inhibitors.

Abbreviations: NSTE-ACS, non-ST-segment elevation acute coronary syndrome; PDE-5, phosphodiesterase type 5.

Source: Modified from M Roffi et al: Eur Heart J 37:267, 2017.

hypotension, or contraindications (e.g., high-degree atrioventricular block, active bronchospasm). Ordinarily, oral beta blockade targeted to a heart rate of 50–60 beats/min is recommended.

Heart rate–slowing calcium channel blockers, e.g., verapamil or diltiazem, are recommended for patients who have persistent symptoms or ECG signs of ischemia after treatment with full-dose nitrates and beta blockers and in patients with contraindications to either class of these agents. Patients who have continuing severe chest pain despite maximal anti-ischemic therapy and are without contraindications to morphine may receive this drug intravenously (1–5 mg every 5–30 min). Additional medical therapy includes angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers. Early administration of intensive HMG-CoA reductase inhibitors (statins), such as atorvastatin 80 mg/d or rosuvastatin 40 mg/d, prior to percutaneous coronary intervention (PCI), and continued thereafter has been shown to reduce both early and late recurrent ischemic events. In patients who do not have an adequate response to maximally tolerated statin (i.e., <50% decrease in low-density lipoprotein cholesterol [LDL-C]), addition of ezetimibe 10 mg daily and/or a PCSK9 inhibitor (alirocumab, evolocumab) early after ACS have been shown to further reduce the LDL-C and prevent future cardiovascular events.

ANTITHROMBOTIC THERAPY

Antithrombotic therapy consisting of antiplatelet and anticoagulant drugs represents the second major cornerstone of treatment **(Table 285-4)**.

Antiplatelet Drugs (See Chap. 123) Initial treatment should begin with the cyclooxygenase inhibitor aspirin with a dose of 150–325 mg of an oral non-enteric-coated preparation or 75–250 mg intravenously. Thereafter, lower oral doses (75–100 mg/d) are recommended since they maintain efficacy while causing less bleeding. Contraindications are severe active bleeding and aspirin allergy.

In the absence of a high risk for bleeding, patients with NSTE-ACS, irrespective of whether an invasive or conservative strategy (see below) is selected, should also receive a platelet $P2Y_{12}$ receptor blocker to inhibit platelet activation. There are now three oral and one intravenous $P2Y_{12}$ inhibitors available. The thienopyridine clopidogrel is an inactive prodrug that is converted into an active metabolite that causes irreversible blockade of the platelet $P2Y_{12}$ receptor. The loading dose of clopidogrel is 300–600 mg, whereas the maintenance dose is 75 mg daily. When clopidogrel is added to aspirin, so-called dual antiplatelet therapy (DAPT), in patients with NSTE-ACS, it confers a 20% relative reduction in cardiovascular death, MI, or stroke, compared to aspirin alone but is associated with a moderate (absolute 1%) increase in major bleeding.

Two other $P2Y_{12}$ inhibitors have been shown to be superior to clopidogrel in preventing recurrent cardiac ischemic events but both increase bleeding. Prasugrel, also a thienopyridine, achieves a more rapid onset and higher level of irreversible platelet inhibition than clopidogrel. Compared to clopidogrel, prasugrel significantly reduced the combined risk of cardiovascular death, MI, stroke, and stent thrombosis, but increased bleeding, in patients with ACS with scheduled PCI. It has been approved for ACS patients following angiography when PCI is planned; it should be administered at a loading dose of 60 mg followed by 10 mg/d. Prasugrel is contraindicated in patients with prior stroke or transient ischemic attack or at high risk for bleeding, and should be administered once the coronary anatomy is known (i.e., post coronary angiography).

Ticagrelor, a potent, *reversible* platelet $P2Y_{12}$ inhibitor, reduces the risk of cardiovascular death, total mortality, or MI compared to clopidogrel across a broad spectrum of patients with ACS. After a loading dose of 180 mg, 90 mg bid is administered as maintenance. Like prasugrel, ticagrelor increases the risk of bleeding. Unlike prasugrel, ticagrelor demonstrated benefit whether patients were managed conservatively or with an early invasive strategy (see below). Some patients may develop dyspnea soon after administration, although the symptoms are often transient and infrequently serious.

Up to one-third of patients have an inadequate response to clopidogrel, and a substantial proportion of these cases are related to a genetic variant of the cytochrome P450 system involving the *2C19* gene that leads to reduced conversion of clopidogrel into its active metabolite. Thus, alternate $P2Y_{12}$ blockers (prasugrel or ticagrelor) should be considered in patients with NSTE-ACS who develop a new coronary event while receiving clopidogrel and aspirin. An intravenous, direct, and rapidly acting $P2Y_{12}$ inhibitor, cangrelor, has also shown benefit relative to clopidogrel in patients who underwent PCI following a NSTE-ACS and reduced the risk of cardiovascular ischemic events, but at the expense of increased major bleeding events.

In the 1990s and early 2000s, several trials of glycoprotein IIb/IIIa inhibitors in patients with NSTE-ACS had shown modest benefit counterbalanced by an increase in major bleeding. However, the majority of earlier studies were performed without concomitant $P2Y_{12}$ inhibitor treatment, and more recent studies in patients receiving the latter failed to show a benefit of routine early initiation of a glycoprotein IIb/IIIa inhibitor. Because of the increased risk of bleeding, the addition of a glycoprotein IIb/IIIa inhibitor to aspirin and a $P2Y_{12}$ inhibitor (i.e., triple antiplatelet therapy) should be reserved for patients undergoing PCI with a high coronary thrombus burden on angiography or thrombotic complications of PCI.

Long-term DAPT should continue preferably for 12 months in patients with NSTE-ACS without an indication for long-term

TABLE 285-4 Clinical Use of Antithrombotic Therapy

Oral Antiplatelet Therapy	
Aspirin	Loading dose of 150–325 mg orally of nonenteric formulation followed by 75–100 mg/d of an enteric or a nonenteric formulation
Clopidogrel	Loading dose of 600 mg (if PCI planned) or 300 mg (if no PCI planned), followed by 75 mg/d
Prasugrel	Pre-PCI: Loading dose of 60 mg followed by 10 mg/d In patient with body weight <60 kg or >75 years old, maintenance dose of 5 mg/d
Ticagrelor	Loading dose of 180 mg followed by 90 mg twice daily
Intravenous Antiplatelet Therapy at the Time of PCI	
Cangrelor	30 μg/kg bolus followed by 4 μg/kg/min infusion for 2 h or duration of procedure
Eptifibatide	180 μg/kg bolus followed in 10 min by second bolus of 180 μg/kg with infusion of 2.0 μg/kg/min for up to 18 h
Tirofiban	25 μg/kg per min over 3 minutes, followed by infusion of 0.15 μg/kg/min for up to 18 h
Parenteral Anticoagulants[a]	
Unfractionated heparin	Bolus 70–100 U/kg (maximum 5000 U) IV followed by infusion of 12–15 U/kg per h titrated to ACT 250–300 s
Enoxaparin	0.5 mg/kg IV bolus at the time of PCI *or* 1 mg/kg subcutaneous every 12 h; the first dose may be preceded by a 30-mg IV bolus; renal adjustment to 1 mg/kg once daily if creatine clearance <30 mL/min
Bivalirudin	Initial IV bolus of 0.75 mg/kg followed by an infusion of 1.75 mg/kg per h
Fondaparinux	2.5 mg subcutaneously daily (only prior to PCI)
Oral Anticoagulant Drugs (concomitant treatment after PCI)	
VKA	Dosing based on INR value and the respective clinical indication
Apixaban	Maintenance dose 5 mg (dose reduced to 2.5 mg) twice daily
Dabigatran	Maintenance dose 150 mg (dose reduced to 75 mg) twice daily in the United States; outside the United States, either 110 or 150 mg twice daily may be used
Edoxaban	Maintenance dose 60 mg (dose reduced to 30 mg) once daily
Rivaroxaban	Maintenance dose 20 mg (dose reduced to 15 mg) once daily In patients without atrial fibrillation or venous thromboembolism 2.5 mg bid may be used

Note: All dose reductions for patients meeting criteria. **(See also Chap. 123.)**

Abbreviations: ACT, activated clotting time; INR, international normalized ratio; PCI, percutaneous coronary intervention; VKA, vitamin K antagonists (e.g., warfarin).

Source: Modified from FJ Neumann: Eur Heart J 40:137, 2019.

full-dose anticoagulation; the duration of DAPT is dependent upon the risk of bleeding versus thrombosis. Clinicians should select the antiplatelet regimen that provides the best balance between efficacy and safety based on the individual patient characteristics and clinical scenario. A shorter duration of DAPT may be considered for patients at high risk of bleeding who are not high at high ischemic risk (see below). Longer duration DAPT, i.e., >12 months, is favored in patients with high atherothrombotic risk (e.g., due to stenting of the left main coronary artery or proximal left anterior descending or proximal bifurcating coronary arteries, recurrent MI, stent thrombosis).

Anticoagulants (See Chap. 123) Four parenteral options are available for anticoagulant therapy to be added to antiplatelet agents: (1) unfractionated heparin (UFH), long the mainstay of therapy; (2) the low-molecular-weight heparin (LMWH) enoxaparin, which has been shown to be superior to UFH in reducing recurrent cardiac events, especially in patients managed by a conservative strategy (however, it is accompanied by a modest increase in bleeding); (3) bivalirudin, a direct thrombin inhibitor that is similar in efficacy to either UFH or LMWH and is used just prior to and/or during PCI; and (4) fondaparinux, a synthetic factor Xa inhibitor that is equivalent in efficacy to enoxaparin but has a lower risk of major bleeding. While UFH and enoxaparin have been widely studied in patients managed either with an early conservative or invasive strategy, bivalirudin is rarely used in conservatively managed patients, while fondaparinux requires supplemental UFH to prevent procedure-related thrombosis. In patients who develop NSTE-ACS while receiving treatment with a direct oral anticoagulant (DOAC), the DOAC **(see Chap. 123)** should be held when one of the four parenteral anticoagulants is begun.

If an early invasive strategy is indicated (see below), radial arterial access is recommended to reduce the risk of bleeding. Excessive bleeding is the most important adverse effect of all antithrombotic agents, including both antiplatelet agents and anticoagulants. Therefore, attention must be directed to the doses of antithrombotic agents, accounting for age, body weight, creatinine clearance, and a previous history of excessive bleeding. Patients who have experienced a stroke are at higher risk of intracranial bleeding with potent antiplatelet agents and combinations of antithrombotic drugs. In patients with atrial fibrillation (including patients with NSTE-ACS) treated with an oral anticoagulant who undergo PCI, the duration of DAPT should be shortened (e.g., stop aspirin after hospital discharge or at most at 1 month after PCI, except in patients at very high risk for ischemic events), and continue $P2Y_{12}$ inhibitor plus DOAC for 1 year. After 1 year, the majority of patients should be transitioned to oral anticoagulation monotherapy without concomitant antiplatelet treatment.

INVASIVE VERSUS CONSERVATIVE STRATEGY (FIG. 285-4)

In an invasive strategy, following initiation of anti-ischemic and antithrombotic agents as described above, coronary arteriography is recommended within 2 h of presentation for patients are unstable or within 24 h in high-risk stable patients, followed by coronary revascularization (PCI or coronary artery bypass grafting), depending on the coronary anatomy (Fig. 285-4). Multiple clinical trials have demonstrated the benefit of this strategy in high-risk patients (i.e., patients with multiple clinical risk factors, ST-segment deviation, and/or positive biomarkers). Two studies comparing an immediate invasive strategy (median time to intervention of 1.4 and 4.7 h after presentation) reduced the rate of death or new MI compared to a delayed invasive strategy (median time of intervention of 61 and 62 h), with a greater benefit among patients with a high-risk score. In patients at low risk, the outcomes from an

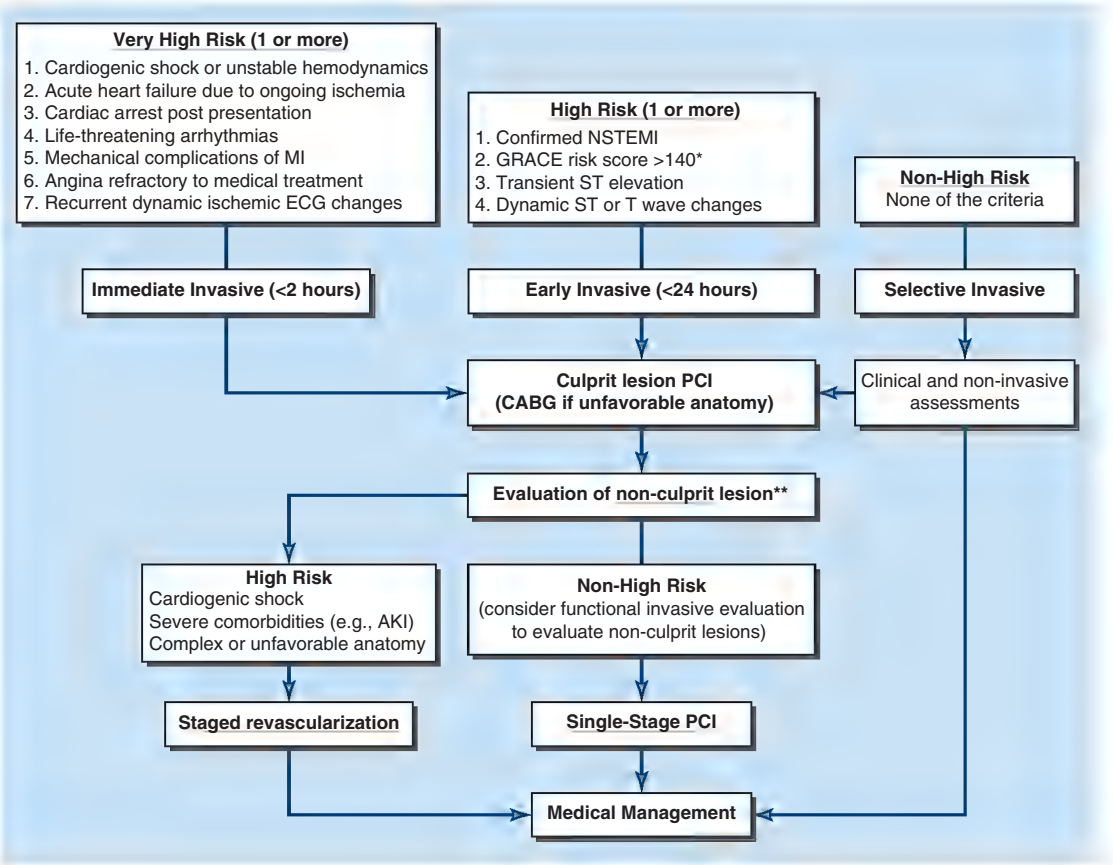

FIGURE 285-4 Selection, timing, and staging of invasive strategy in patients with non-ST-elevation acute coronary syndrome. AKI, acute kidney injury; CABG, coronary artery bypass grafting; GRACE, Global Registry of Acute Coronary Syndrome Events; PCI, percutaneous coronary intervention. Patients with non-ST-elevations acute coronary syndrome who have one or more very-high risk features should undergo immediate invasive strategy with coronary angiography as soon as possible (within 2 hours), unless absolute contraindications exist. An early invasive strategy should be considered within 24 hours in stable patients with one or more high risk features. Stable patients with no very-high or high risk features should be managed with a selective invasive approach in which clinical, non-invasive assessments, and patient preferences are used to guide management. *Fox KA, BMJ Open 2014;4:e004425. https://doi.org/10.1136/bmjopen-2013-004425. **In patients with multivessel disease, single-stage PCI during the index procedure to achieve complete coronary revascularization should be considered in the absence of cardiogenic shock, severe comorbidities such as AKI, or coronary anatomy that is unfavorable or high risk requiring evaluation for The Heart Team for CABG vs complex coronary intervention.

invasive strategy are similar to those obtained from a conservative strategy. The latter consists of anti-ischemic and antithrombotic therapy followed by a "selective invasive approach," in which the patient is observed closely and coronary arteriography is conducted if coronary computed angiography shows the presence of epicardial coronary stenosis, rest pain or ST-segment changes recur, a biomarker of necrosis becomes positive, or there is evidence of severe ischemia on a stress test.

LONG-TERM MANAGEMENT

The time of hospital discharge is a "teachable moment" for the patient with NSTE-ACS, when the caregiver can review and optimize the medical regimen. Risk factor modification is crucial, and the importance of smoking cessation, following an appropriate diet, achieving and maintaining optimal weight, daily exercise, an LDL-C <55 mg/dL, blood pressure control, and control of hyperglycemia should be emphasized. There is evidence of benefit with long-term therapy with several classes of drugs. Beta blockers, intensive lipid-lowering therapies (high-intensity statin, ezetimibe, and if needed, adding a PCSK9 inhibitor or bempedoic acid) to achieve an LDL-C <55 mg/dL, ACE inhibitors or angiotensin receptor blockers, and sodium-glucose cotransporter 2 or glucagon-like peptide 1 receptor agonists in selected patients with type 2 diabetes mellitus **(see Chap. 416)** are recommended. The recommended antiplatelet regimen consists of the combination of low-dose (75–100 mg/d) aspirin and a $P2Y_{12}$ inhibitor (prasugrel or ticagrelor is favored over clopidogrel) for 12 months unless there is a high risk of bleeding. After 12 months, antiplatelet monotherapy with a single antiplatelet should be continued thereafter **(see Chap. 284)**, unless long-term full-dose anticoagulation is indicated, in which case anticoagulant without antiplatelet therapy is recommended long-term (see above).

Several trials have explored shorter courses and less intensive DAPT in patients not on anticoagulation who are at high risk of bleeding and low risk of ischemic events. Recent guidelines support a shorter-duration DAPT regimens of 3–6 months in these patients, followed by stopping aspirin and continuing with $P2Y_{12}$ inhibitor alone. Another option is de-escalation of $P2Y_{12}$ intensity (i.e., switching from prasugrel or ticagrelor to clopidogrel at 3 months) while continuing aspirin. De-escalation of DAPT is not recommended in the first 30 days following ACS as this has been associated with increased risk of ischemic events.

In patients who require long-term anticoagulation (e.g., patients with atrial fibrillation), the duration of so-called "triple therapy" consisting of aspirin, clopidogrel, and an oral anticoagulant should be kept to a minimum. In these patients, aspirin should be discontinued at discharge or 1 week, and oral anticoagulation and clopidogrel continued until month 12, followed thereafter by oral anticoagulant alone. Conversely, in patients at high ischemic risk (e.g., those with prior MI, diabetes mellitus, coronary vein graft, heart failure) who are also at low risk of bleeding and not on an anticoagulant, continuation of DAPT for up to 3 years has been shown to be beneficial. If ticagrelor is used, the dose should be reduced to 60 mg twice daily 1 year after ACS as this reduces bleeding without an increase in ischemic events.

Registries have shown that women and racial minorities, as well as patients with NSTE-ACS at high risk, including the elderly and patients with diabetes or chronic kidney disease, are less likely to receive evidence-based pharmacologic and interventional therapies with resultant poorer clinical outcomes and quality of life. Special attention should be directed to these groups.

VASOSPASTIC ANGINA

In 1959, Prinzmetal and colleagues described a syndrome of severe ischemic pain that usually occurs at rest and is associated with ST-segment elevation originally known as Prinzmetal's variant angina, but since renamed vasospastic angina (VA). VA is caused by focal spasm of an epicardial coronary artery with resultant transmural ischemia and abnormalities in left ventricular function that may lead to acute MI, ventricular tachycardia or fibrillation, and sudden cardiac death. The cause of the spasm is not well defined, but it may be related to hypercontractility of coronary arterial smooth muscle due to adrenergic vasoconstrictors, leukotrienes, or serotonin. For reasons that are not clear, the prevalence of VA has decreased substantially during the past few decades, although it remains much more frequent in Japan than in North America or Western Europe.

Clinical and Angiographic Manifestations Patients with VA are generally younger and, except for cigarette smoking, have fewer coronary risk factors than do patients with NSTE-ACS. Cardiac examination is usually unremarkable in the absence of ischemia. However, a minority of patients have a generalized vasospastic disorder associated with migraine and/or Raynaud's phenomenon. The clinical diagnosis of VA is made by the detection of transient ST-segment *elevation* with rest pain, although many patients may also exhibit episodes of silent ischemia.

Coronary angiography demonstrates transient coronary spasm as the diagnostic hallmark of VA. Atherosclerotic plaques in at least one proximal coronary artery occur in about half of patients. Hyperventilation and intracoronary acetylcholine have been used to provoke focal coronary stenosis on angiography or to provoke rest angina with ST-segment elevation to establish the diagnosis. In patients with no obstructive coronary atherosclerosis and suspected coronary vasomotor abnormalities, a positive provocative test for spasm has been shown to be safe, identifies a high-risk subgroup, and is endorsed by guidelines since it permits selection of therapy most appropriate for the underlying pathophysiology.

TREATMENT

Vasospastic Angina

Calcium channel blockers are the main therapeutic agents. Nitroglycerin, either sublingual or intravenous, is effective to relieve coronary arterial spasm and angina, but the efficacy of long-term nitrates is limited by nitrate tolerance. Aspirin may actually increase the severity of ischemic episodes, possibly as a result of the sensitivity of coronary tone to modest changes in the synthesis of prostacyclin, and should be used only in patients with significant atherosclerosis. Statin therapy has been shown to reduce the risk of major adverse events, although the precise mechanism is not established. The response to beta blockers is variable. Cilostazol and nicorandil (not available in the United States) may also prevent recurrent episodes. Coronary revascularization may be helpful in patients who also have discrete, flow-limiting, proximal fixed obstructive lesions. Patients who have had ischemia-associated ventricular fibrillation despite maximal medical therapy should receive an implantable cardioverter-defibrillator.

Prognosis Many patients with VA pass through an acute, active phase, with frequent episodes of angina and cardiac events during the first 6 months after presentation, after which there may be a tendency for symptoms and cardiac events to diminish over time. Survival at 5 years is excellent (~90–95%), but as many as 20% of patients experience an MI. Patients with no or mild fixed coronary obstruction experience a lower incidence of cardiac death or MI compared to patients with associated severe obstructive lesions. Patients with VA who develop serious arrhythmias during spontaneous episodes of pain are at a higher risk for sudden cardiac death. In most patients who survive an infarction or the initial 3- to 6-month period of frequent episodes, there is a tendency for symptoms and cardiac events to diminish over time.

GLOBAL CONSIDERATIONS

Ischemic heart disease (IHD) and its most dangerous manifestation, ACS, remain the most frequent cause of death and disability worldwide. In the mid-twentieth century, these conditions were most common in high-income countries. Although improvements in primary prevention and treatment of ACS have reduced cardiovascular morbidity and mortality due to IHD over the past 70 years, these advances have not affected all population groups equally. In Europe, there remains a northeast to southwest gradient, with higher prevalence in northern Russia and the Baltic nations and considerably lower prevalence in Spain, France, and Italy. In the United States, there remain racial and economic disparities, with poorer outcomes in minorities and low-income populations.